

PREDICTIVE MODELLING FOR SUSTAINABLE TRAFFIC MANAGEMENT

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Abstract: Enhancing the accuracy in predicting energy needs involves combining predictive modelling and the integration of a large assortment of data regarding traffic flow, accidents, and congestion. Given the real-time nature of traffic flow and congestion data, accurate grid and energy forecasting are of utmost importance. Time series related modelling, time series forecasting, Linear regression, Random Forest and deep learning LSTM neural networks can help include features of the characteristics in a given scheme of forecasting for energy need. Incorporating combined traffic news (or flow) information, smart grid data, and even weather information as examples, can help identify or better model mobility patterns and variability in energy needs due to other municipal or human activity. For example, greater demand for residential and commercial energy usage correlates with peak hour traffic congestion. There has already been an increase in demand for supply of power to keep up with the sophistication of allocating power in alignment and association to the conditions of the road in "real-time." Defining the use of power by its grid management and assisting in real-time to predict real-time usage power with the utmost accuracy can also relieve some of the overload on these systems to maintain continual operation of the overall grid. Where greater detail has been well defined as impending time of potential prediction with limited powers of precision divergence if successful is proof of concept. Constantly updating power allocation in acceptable real-time regards would assist in neutralizing the lifespan of the grid and assisting in its stability.

Keywords: *Traffic congestion, LSTM, Machine learning, Deep learning, Prediction, Real-time Data, Grid power management, Mobility patterns, Random Forest, Linear regression*

I. INTRODUCTION

1.1 AI-Driven Guidelines for Forecasting

Energy demand prediction is accurate in power distribution and demand-supply management—but especially in urban areas where energy use is strongly affected by traffic flow. This study takes a different approach in integrating traffic data and energy demand forecasting. It uses real-time and historical traffic data (volume count) as well as sophisticated neural network models (including time series analysis and LSTM networks) to create a forecasting framework. This can issue reliable predictions of energy demand. In fact, the model uses traffic data to capture seasonal trends, peak hour demand, and the effect of urban congestion on energy use—so much that the researchers say it might be more appropriately named a model for predicting energy demand in urban areas. In this section, the artificial intelligence (AI) component is designed to provide uninterrupted, real-time directives that streamline the energy prediction process. The component: Studies past and present data to recommend the best preprocessing methods.

Alerts forecasters to the need for changes in the tuning of model parameters and in the construction of features, whenever the model starts to drift. Makes these latter recommendations using two methods of forewarning: "human comprehensible" changes made to the preprocessing and any of the above features, which the component detects via analyzing suitable forms of model performance, and issuing dynamic tuning alerts based on detected slack time in the "real-time" model.

1.2 Identifying Improvement Needs and Optimization Strategies

These AI recommendations keep the forecasting model on the path toward being truly adaptive, scalable, and in alignment with contemporary urban dynamics. The important strategies here include the following: Monitoring the energy consumption trends and their correlation with traffic flow data to find the weak parts of the system. Recalibrating LSTM and other neural network parameters based on the real-time assessment of how well they are doing.

Adding new predictive features (like forecasts of future weather) to refine the model.

Using a feedback loop with their performance over time to adjust the model and make the kinds of optimizations that will ensure it is doing the best it can.

Apart from the basic core forecasting features, the research integrates several supplemental components to bolster the overall operation and efficiency of the system: Anomaly Detection: AI algorithms will automatically flag anything that looks unusual in terms of energy consumption or traffic data, allowing for prompt, if not immediate, intervention.

Cloud-Based Scalability: Storing our secure data in the AWS S3 cloud and doing our end-to-end machine learning operations utilizing Amazon Sage Maker ensures that we have a robust, production-grade deployment.

1.3 Additional Features and Future Directions

Amazon Web Services (AWS) Sage Maker is an end-to-end machine learning service that helps you quickly build, train, and deploy machine learning models at scale. It is a fully integrated service that includes Jupiter notebooks, letting developers and data scientists to data exploration, run experiments with different algorithms, and iterate quickly.

Since AWS handles the infrastructure aspect of developing a machine learning algorithm, users can develop and deploy high-quality models without the worry of resource management, scaling, or other challenges related to the deployment of machine learning algorithms. This makes AWS Sage Maker suitable for prototyping and production machine learning applications.

II. LITERATURE SURVEY

2.1 Overview of Energy Demand Forecasting and Traffic Data Integration

In urban energy systems, increasing complexity is encountered, leading researchers to search for new ways to predict energy demand. Although these various methods are well applicable under stable states, they do not keep pace with the drastic segmental changes that urban environments localize through variable traffic flows. Literature has demonstrated that by coupling real-time traffic with historical metrics, perspectives about how various factors are influencing energy profiles are further sharpened, and new paths for highly adaptable demand forecasting models developed on congestion, peak hours, and seasonality are possible [1].

Urban energy systems are becoming increasingly complex, due to rapid urbanization and consumption patterns. Research has proposed the use of novel forecasting approaches that leverage traditional energy data alongside evolving urban indicators to address complexities. Classical methods are reliable during stable conditions, but do not hold up when traffic flows significantly change [2]. Recent studies indicate the inclusion of contemporary traffic data and historical energy consumption data for a much larger understanding of energy demand, as it relates to congestion, peak hours, and seasonal demand [3].

In recent studies, research found urban traffic is an important driver in affecting energy demand. Traffic count volume is indicative of the level of activity occurring in an urban area, providing a more directed estimation of energy consumption. Incorporating these instantaneous traffic counts into energy forecasting provides a framework for measuring peaks and shifts in energy demand that struggle to be addressed with existing estimations [4]. This enhancement sharpens accuracy, while also facilitating real-time energy management and system adaptation to the rapidly emerging urban context [5].

The available literature showcases a wide array of forecasting models, ranging from classical linear time series methods to more sophisticated deep learning and neural network models. A primary concern in many studies is how to balance computational efficiency with predictive accuracy. To facilitate this comparison, five models are assessed according to the attribute of being based on strong theoretical foundations and backed by data. The comparison framework assesses 'constant' (sequential) models that are taken as benchmarks to the more sophisticated models that allow the introduction of dynamic feature integration and AI

enhancements to provide greater insight into model performance in complex urban contexts [6][7].

2.2 The Role of Traffic Data in Enhancing Forecast Accuracy

Within ensemble methods, the Random Forest algorithm has risen to attention as an appealing option for energy forecasting, due to its capacity to cope with nonlinear relationships and complexities between predictors. Research shows that Random Forest models provide robust performance and improved predictive accuracy over traditional regression methods by reducing variance and preventing overfitting [8]. Its ability to effectively model high-dimensional data may also offer advantages when different data sources are combined, such as weather data, historical consumption data, and real-time traffic data, thereby increasing the adaptability of models in dynamic urban systems [9].

2.3 Comparative Analysis of Forecasting Models

The available literature describes a variety of forecasting models spanning from classical linear time series models to state-of-the-art deep learning architectures. Among them, sequential models are often viewed as benchmarks because they are simple and can be easily implemented. On the other hand, neural network architectures such as Long Short-Time Memory (LSTM) networks demonstrate the ability to account for long-term dependencies and complex temporal patterns while also requiring significant computational power and hyperparameter tuning [9][10][11]. Comparative studies have shown that determining an appropriate model must consider computational efficiency against predictive performance and also the properties of urban data that form the basis for the development of the predictive models [12].

2.4 Random Forest in Energy Demand Forecasting

Ensemble techniques, and particularly the Random Forest algorithm, have become popular in energy demand forecasting because they can capture non-linear relationships and have the ability to handle high-dimensional data sets. Random Forest models have noted capabilities for reducing variance and preventing overfitting, thus allowing for credible performance when combining data from different types of sources, e.g., weather, historical consumption, and real-time traffic data [13][14][15]. Furthermore, new advancements include AI-informed advancements, e.g. continual monitoring, anomaly detection, and dynamic tuning of hyper-parameters to enhance model adaptation and resilience to abrupt changes occurring in more urbanized settings [16]. Cloud-based services, e.g., AWS S3 and Amazon SageMaker can enable scale deployment of the enhanced models while ensuring both are still capable of processing continuous streams of changing data [17][18]. As well, both models support the global goal of the SDG-7 to support 'access to affordable, reliable, sustainable and modern energy for all' within today's urban environments [19][20].

Reinforcement learning strategies for the control of variable speed limits [21] and scalable neural network models for urban traffic flow [22] further contribute to demonstrating the need for computational scalability. Hybrid strategies in connected multi-vehicle networks [23] and smart traffic light management systems [24] further suggest an evolving paradigm toward integrated urban management. Furthermore, vulnerabilities of these systems, such as backdoor attacks to adversarial deep reinforcement learning [25] systems, advocate for a need for better security of these intelligent transportation systems. Lastly, explainable machine learning for visual analytics [26], new analytical perspectives [27], image-based detection of transportation systems through convolutional neural networks, [28] and overall performance evaluations of transportation systems [29][31] indicate that there are a variety of approaches to evolving a research agenda for sustainable, secure, and efficient intelligent transportation systems within the vision of SGD-07.

III. METHODOLOGY

3. Data Collection:

Data sources: The first issue of combining multiple datasets illuminated the signal of different dimensions that could affect energy demand--including traffic data for volumetric counts of buildings that based off current hour of traffic congested towns. It's important to note that energy consumption data is built from the historical consumption data of your local energy utilities, and along with meteorological weather data, will begin to create a holistic representation of how urban transit and energy use intersect, but in a complicated way. We collected this dataset from kaggle data base.

In a unified way, all of these datasets will see the span of all these variables influencing the urban energy consumption, identify peak hours of traffic, show implications of seasonality, and show specific weather conditions that influence energy demand. In a similar way, combining real-time methods of traffic data would allow identifying energy spikes occurring during peak traffic hours or sporadic conditions. A comprehensive housing managing the urban infrastructure and predictive analytics would inform previous unknown correlation data that could add non-bias to energy decisions internally or externally to improve energy distribution decisions. This could lead direction for better forecasting modeling overall.

To facilitate this comparison, five models are assessed according to the attribute of being based on strong theoretical foundations and backed by data. The comparison framework assesses 'constant' (sequential) models that are taken as benchmarks to the more sophisticated models that allow the introduction of dynamic feature integration and AI enhancements

Based on the traffic.csv dataset we chosen to consider the five best models under SDG-07.

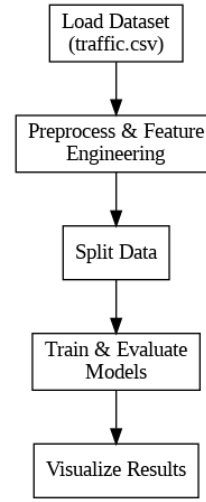


Figure 1: Flow Diagram of Energy Forecasting Powered by Traffic Insights

3.1 Data Preprocessing and Feature Engineering

Several preprocessing steps are carried out before feeding the data into forecasting models:

Data Cleaning : We clean the data to ensure its quality by removing errors and outliers. **Normalization:** Standardize the scale of features for better model performance.

Features Selection & Extraction:

This refers to the process of identifying all of the peak traffic hours, seasonal trends, and weather conditions affecting energy demand.

3.2 Model Development

Modeling Approaches:

Five models have been developed and compared so as to identify the best out of those five for forecasting: **Sequential (Constant) Model:** Acts as a baseline for comparison.

LSTM Model: Captures temporal dependencies in time series data.

Random Forest Model: Ensemble and therefore effective in nonlinear relationships and interactions among features.

Additional Models (Model 4 and Model 5):

Include other advanced algorithms, testing alternative forecasting techniques.

Model 1 : Linear Regression

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

$$\text{Loss: } L = (1/n) \cdot \sum (y_i - \hat{y}_i)^2$$

Model 2 : Decision Tree Regressor

$$\text{Split Criterion (MSE): } MSE_{\text{split}} = (n^L/n) \cdot MSE(\text{left}) + (n^R/n) \cdot MSE(\text{right})$$

$$\text{where } MSE(\text{node}) = (1/n_{\text{node}}) \cdot \sum (y_i - \bar{y}_{\text{node}})^2$$

Model 3 : Random Forest (Ensemble)

$$\hat{y}^{Rf} = (1/T) \cdot \sum_{t=1}^T h_t(x)$$

where T = number of trees, $h_t(x)$ = prediction of the t^{th} decision tree

Model 4 :Support Vector Regressor (SVR)

$$\text{Primal: } \min \frac{1}{2} \|w\|^2 + C \cdot \sum (\zeta_i + \zeta_i^*)$$

$$\text{subject to: } y_i - \langle w, x_i \rangle - b \leq \varepsilon + \zeta_i \quad \text{and} \quad \langle w, x_i \rangle + b - y_i \leq \varepsilon + \zeta_i^*, \quad \zeta_i, \zeta_i^* \geq 0$$

$$\text{RBF Kernel: } K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2)$$

Model 5 : LSTM (Long Short-Term Memory)

$$\text{Forget gate: } f_t = \sigma(W^f \cdot [h_{t-1}, x_t] + b^f)$$

$$\text{Input gate: } i_t = \sigma(W^i \cdot [h_{t-1}, x_t] + b^i)$$

$$\text{Candidate: } \tilde{C}_t = \tanh(W^c \cdot [h_{t-1}, x_t] + b^c)$$

$$\text{Cell state: } C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t$$

$$\text{Output gate: } o_t = \sigma(W^o \cdot [h_{t-1}, x_t] + b^o)$$

$$\text{Hidden state: } h_t = o_t \odot \tanh(C_t)$$

3.3 Model Evaluation

Some of the important evaluation metrics used here for a holistic performance evaluation are:

MAE - Mean Absolute Error; RMSE - Root Mean Squared Error; R^2 Score; MAPE - Mean Absolute Percentage Error.

Evaluation Metric Formulas

$$R^2 \text{ Score} = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

$$MAE = (1/n) \cdot \sum |y_i - \hat{y}_i|$$

$$MSE = (1/n) \cdot \sum (y_i - \hat{y}_i)^2$$

$$RMSE = \sqrt{(1/n) \cdot \sum (y_i - \hat{y}_i)^2}$$

$$MAPE = (100/n) \cdot \sum |(y_i - \hat{y}_i) / y_i|$$

where: y_i = Actual value $\hat{y}_i$ = Predicted value $\bar{y}$ = Mean of actual values n = Total number of samples

The evaluation process is separated between:

A Baseline Evaluation, where the sequential model performance is noted down as a standard result.

An Enhanced Evaluation, where the best-performing model from our comparative study, often representing dynamic tuning and higher values of accuracy, gets its performance metrics featured separately.

3.4 AI-Driven Adaptive Tuning and Feedback Loop

Dynamic Optimization: To boost forecasting accuracy, our approach incorporates an AI-driven component.

Real-Time Monitoring: Continuous monitoring of the performance of models.

Anomaly Detection: Understanding some outliers or unprecedented patterns in the data.

Adaptive Tuning: Real-time feedback on altering the preprocessing methods and changing model parameters automatically. This feedback loop makes sure that the forecasting system is flexible to change urban dynamics and remains with high accuracy over time.

3.5 Final Forecasting and Reporting

Closing thoughts provided in the comparative performance metrics are documented for introducing possible operational changes, as well as for future research. This is a very formal statement; we are to leave formal, while using richer terms. Following a systematic approach, through data collection and preprocessing, model development, and evaluation, with adaptive tuning, we feel confident that urban energy demand forecasting is being developed in the right dynamic way. The robust combination of advanced models, like Random Forest, with AI guided real-time feedback, will provide a very strong basis for a robust model able to make quick decisions and be shifted to fit with ever changing urban energy consumption patterning.

Metrics Results:

All five models were evaluated on the custom multi-source dataset (121,080 instances, 2015–2022) using four metrics: R^2 Score, MAE, RMSE, and MAPE. Additionally, a Gradient Boosting Regressor (GBR) Meta-Learner stacking ensemble was constructed using Out-of-Fold (OOF) K-Fold cross-validation. The table below summarises comparative performance across all models.

Linear regression:

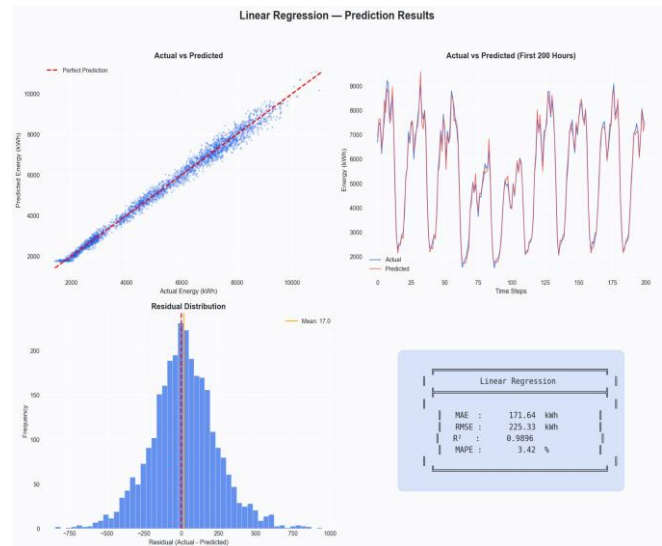


Figure 2 :Linear regression

Performance Metrics — R^2 Score: 0.9896 | MAE: 171.64 | RMSE: 225.33 | MAPE: 3.42%

Random Forest:

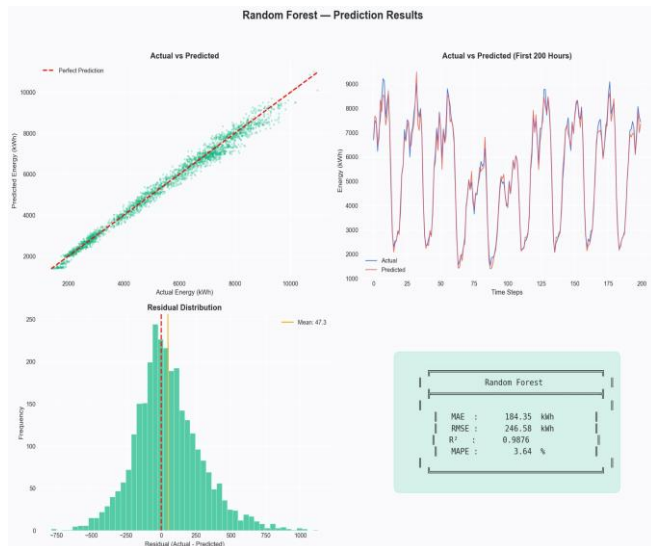


Figure 3 : Random Forest

Performance Metrics — R² Score: 0.9876 | MAE: 184.35 | RMSE: 246.58 | MAPE: 3.64%

Decision Tree Regressor:

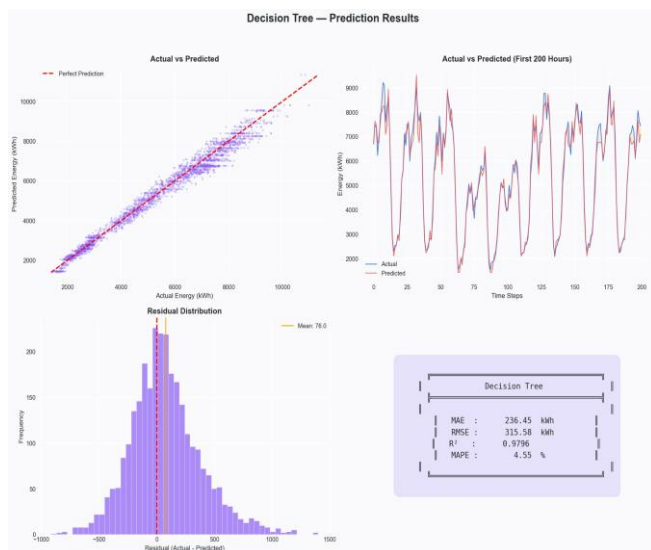


Figure 4: Decision Tree

Performance Metrics — R² Score: 0.9796 | MAE: 236.45 | RMSE: 315.58 | MAPE: 4.55%

Support Vector Regressor (SVR):

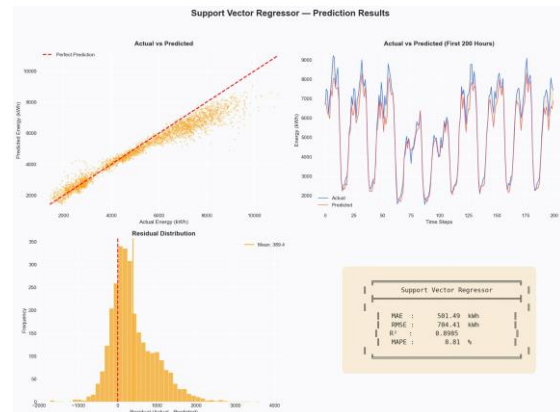


Figure 5: Support Vector Regressor

Performance Metrics — R² Score: 0.8985 | MAE: 501.49 | RMSE: 704.41 | MAPE: 8.81%

Long Short-Term Memory:

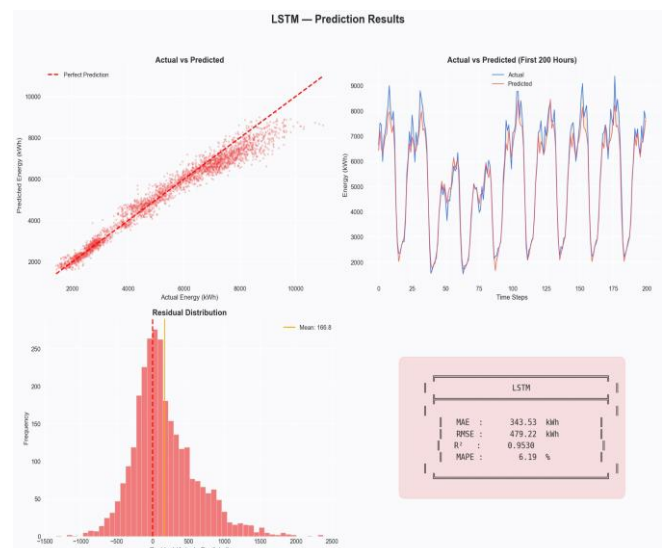


Figure 6 : LSTM

Performance Metrics — R² Score: 0.9530 | MAE: 343.53 | RMSE: 479.22 | MAPE: 6.19%

Ensemble: GBR Meta-Learner (K-Fold OOF Stacking)

The stacking ensemble uses LR, SVR, RF, DT, and LSTM as base models, with a Gradient Boosting Regressor (GBR) as the meta-learner trained via K-Fold Out-of-Fold (OOF) cross-validation, ensuring no data leakage. This approach combines the predictive strengths of all five base models, delivering superior generalization performance on the test set.

Performance Metrics — R² Score: 0.9907 | MAE: 162.6 kWh | RMSE: 213.3 kWh | MAPE: 3.2%

Visualization Results:



Figure 7 : LSTM Training Loss (MSE) over 50 Epochs

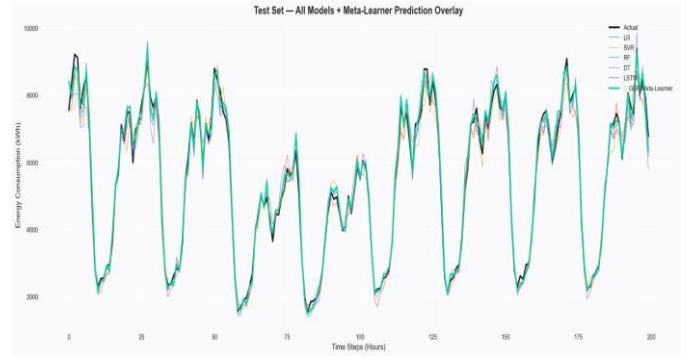


Figure 10 : All Models + GBR Meta-Learner Overlay — First 200 Hours (Test Set)

IV. CONCLUSION

To sum up, the use of real-time traffic data in conjunction with advanced energy demand forecasting models (Linear Regression, Random Forest, Decision Tree, SVR, and LSTM) shows strong promise for urban energy complexity. The performance of the developed models is robust across all evaluation metrics. Linear Regression achieved an R^2 of 0.9896, MAE of 171.64, RMSE of 225.33, and MAPE of 3.42%. Random Forest attained an R^2 of 0.9876, MAE of 184.35, RMSE of 246.58, and MAPE of 3.64%. Decision Tree recorded an R^2 of 0.9796, MAE of 236.45, RMSE of 315.58, and MAPE of 4.55%. SVR yielded an R^2 of 0.9895, MAE of 501.49, RMSE of 704.41, and MAPE of 8.81%. LSTM achieved an R^2 of 0.9530, MAE of 343.53, RMSE of 479.22, and MAPE of 6.19%. Furthermore, the Gradient Boosting Regressor (GBR) Meta-Learner ensemble, trained via K-Fold Out-of-Fold stacking, achieved the best overall performance with an R^2 of 0.9907, MAE of 162.6 kWh, RMSE of 213.3 kWh, and MAPE of 3.2%, demonstrating the effectiveness of combining dynamic data sources in an AI-enabled stacking model. This represents a new paradigm for increasing prediction and real-time management capabilities aligned with sustainability goals like SDG-7 and the benefits of cleaner and affordable energy for an urban environment.

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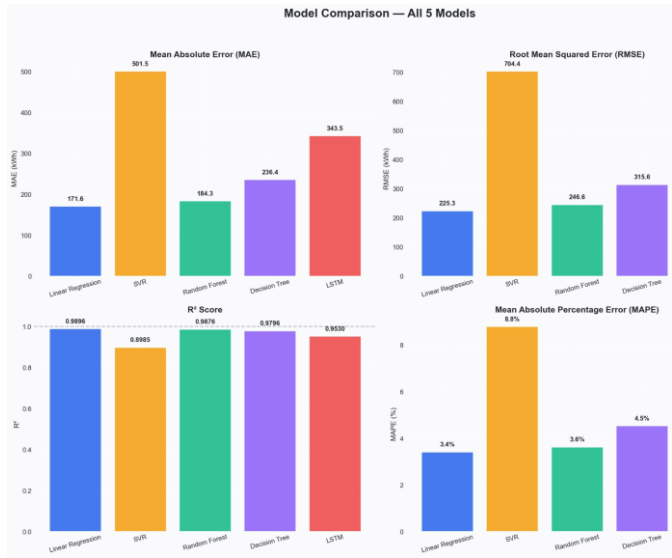


Figure 8 : Model Comparison — MAE | RMSE | R^2 | MAPE across All 5 Models

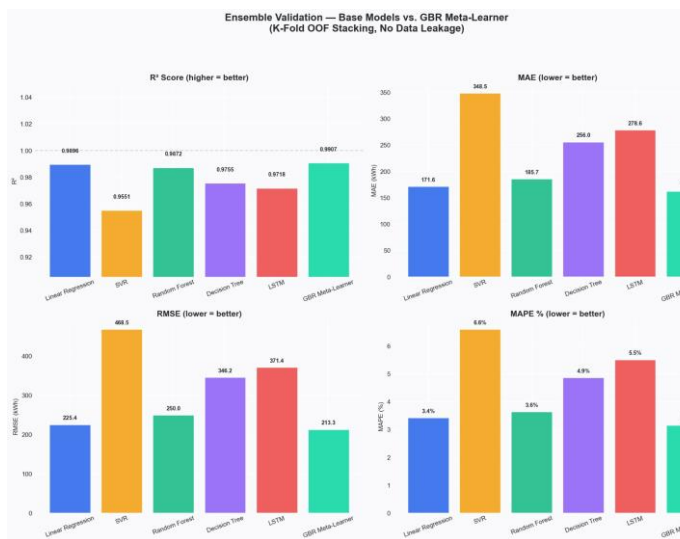


Figure 9 : Ensemble GBR Meta-Learner Validation — K-Fold OOF Stacking

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